LESSON 3

JUMP DIFFUSION MODELS

Derivative Pricing - Module 7

WORLDQUANT UNIVERSITY

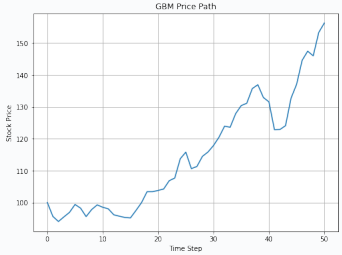
Outline

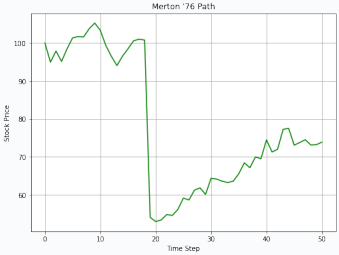
- Jump diffusion models
- Merton (1976) model
- Poisson processes

Jump diffusion models

We started with a **normally distributed world** (e.g., Black-Scholes), and moved to one in which **volatility is non-constant** (e.g., CEV) and **stochastic** (e.g., Heston). Our focus has been primarily on the volatility part of the underlying price process. Let's now turn to the **drift component**.

- The GBM model (even in Heston) produces an arguably "smooth" stock price evolution.
- However, this is not really what happens in reality, where prices constantly experience jumps, e.g.,:
  - Due to good (e.g., earnings surprise) or bad (e.g., profit warning) **fundamental reasons**.





Merton (1976) model

One of the most famous jump diffusion models for option pricing is **Merton (1976)**:

Merton, Robert C. "Option pricing when underlying stock returns are discontinuous." Journal of Financial Economics, vol. 3, no. 1–2, 1976, pp. 125–144.

Merton's '76 model SDE includes both a Brownian motion and a Poisson jump process with intensity λ and average jump size μ_j:

$$dS_t = \left(r - r_j\right) S_t dt + \sigma S_t dZ_t + J_t S_t dN_t$$

where,

- $r_j = \lambda \left(e^{\mu_j + \frac{\delta^2}{2}} \right) - 1$

  is the "correction" to the drift term in order to maintain the risk-neutral measure.

- J_t is a jump at time t, which follows the distribution:

$$Log(1 + J_t) \approx Normal \left(log(1 + \mu_j) - \frac{\delta^2}{2},\ \delta^2 \right)$$

Merton (1976) model

Similar to what we have done with other previous models, we can use Monte Carlo methods on the discretized version of the model for pricing:

$$S_t = S_{t-1}\left(e^{\left(r - r_j - \frac{\sigma^2}{2}\right)dt + \sigma\sqrt{dt}z_t^1} + \left(e^{\mu_j + \delta z_t^2} - 1\right) y_t\right)$$

Note that we have **3 sources of randomness** here:

1. Stock price diffusion (standard normal): z_t^1
2. Size of the jump (standard normal): z_t^2
3. Timing of the jump (Poisson): y_t

Also, note the importance of variables μ_j and λ, the average size and intensity (frequency) of the jump:

- How to assess these parameters? $\rightarrow$ Calibration
- In the next lesson, we will implement the model and play around with these.

Poisson processes

Merton (1976) models stock price jumps in a very convenient way using random processes following a **Poisson distribution**.

If you are familiar with Poisson distributions, this should not surprise you, as it is a process often used to model the **occurrence of events**. Some applications are:

- Traffic flow $\rightarrow$ theory of queues, how long should I wait for the bus?
- Firm bankruptcy $\rightarrow$ assess the probability of default.
- Insurance claims $\rightarrow$ probability of reporting an accident & assessment of insurance premium.

The probability mass function of a Poisson-distributed random variable X is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where k is the number of occurrences of the event, and $\lambda > 0$.

$\Rightarrow$ Also, note that $\lambda = E(X) = Var(X)$

Summary of Lesson 3

In Lesson 3, we have looked at:

- Jump diffusion models as a way to get closer to real-life behavior of prices.
- Merton (1976) jump diffusion model.
- Poisson process basics and its applications to finance.

⇒ **TO DO NEXT**: This lesson does not have an associated Jupyter Notebook. Jump to the next lesson to see a complete implementation of the Merton (1976) model in Python, together with a discussion of the distribution that it generates for stock prices/returns and the parameters of the model.

⇒ In the next lesson, you will learn how to implement the Merton (1976) model in Python.